

From one image to a production-ready interactive 3D asset

Steampunk Ballista Turret | Vite + TypeScript + Three.js | Procedural geometry, mechanics, QA and deployment

REFERENCE — IMG2THREEJS GATES — MANUAL V2 REFINEMENT — AUTOMATED QA — GITHUB + RAILWAY



INPUT / SINGLE STYLIZED REFERENCE



OUTPUT / LIVE INTERACTIVE BROWSER ASSET

EXECUTIVE SUMMARY

Starting from one stylized render, we built a complete code-generated 3D asset and inspection experience. The final system includes mechanical state, semantic metadata, export, automated QA, and a live public deployment.

OUTCOME

Code-only model. No runtime secrets. No third-party 3D asset. Production URL returns HTTP 200.

245

MESH INSTANCES

16,062

TRIANGLE FACES

28

SEMANTIC NODES

36/36

UNIT TESTS

3/3

BROWSER JOURNEYS

1

PRODUCTION DEPLOY

01 What we engineered

- Procedural geometry and PBR-style materials authored in TypeScript.
- 28-node semantic hierarchy, four sockets, colliders and destruction groups.
- Real yaw and pitch pivots, raycast selection and exploded view.
- Browser-native binary GLB export and a responsive control surface.

02 Critical refinements

- Corrected bow sweep and tangent-aligned endpoint clamps.
- Closed the pod underside, rebuilt chains and added bilateral fasteners.
- Separated Fire from Crank & Load with an explicit four-state cycle.
- Centered pitch and yaw, completed Reset and removed the conflicting scope.

03 Production delivery

- 36 Vitest tests plus three Playwright interaction and visual journeys.
- GitHub Actions runs typecheck, build, Chromium and evidence retention.
- Railway uses one healthy deployment, healthchecks and path-filtered updates.
- Dark and light desktop, 390 px mobile, reference modal and GLB verified.

TRUST BOUNDARY

Single-image reconstruction: hidden geometry is an informed approximation. Projectile travel is deterministic animation, not rigid-body physics. The GLB omits procedural roughness DataTextures. No database, API key or runtime secret is required.

LIVE <https://ballista-production.up.railway.app>

SOURCE <https://github.com/edemaistre/steampunk-ballista-turret>

